Seeing a switching device: measurements and evidence

In a nanoscale device, electrical response is a measured outcome; a filament, its composition and its motion are explanations that require additional evidence. This chapter connects an experiment, its signal and the limits of the conclusion.

Numerical examples are hypothetical. Images are original schematics; experimental figures and published performance values are not reproduced.

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Start with an evidence chain

Ask four questions in order: **What was driven? What was measured? How was it calibrated? Which mechanisms remain compatible with the data?**

An I–V change shows an electrical state change under the stated protocol. A time-correlated image supplies structural evidence; chemical analysis and control measurements can narrow the interpretation. No single contrast feature automatically establishes composition, valence, or a unique atomic mechanism.

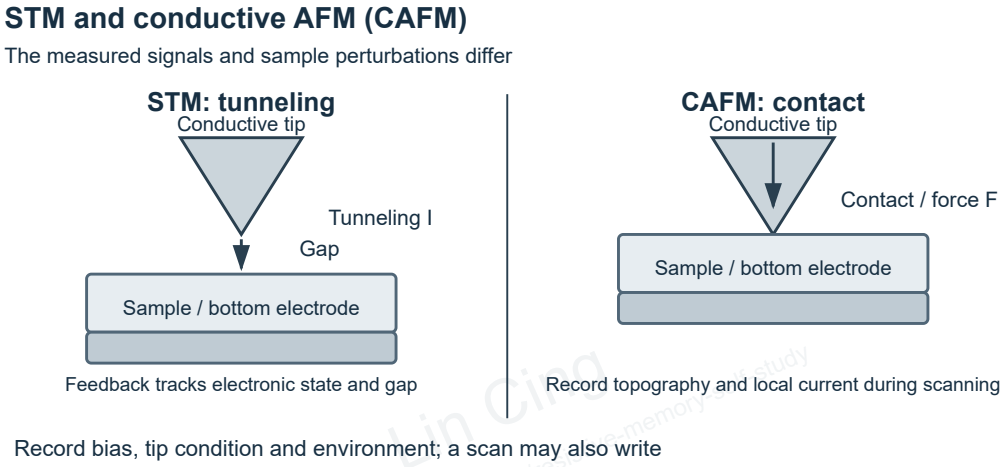
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Choose the probe for the question

Method	Primary signal	Useful question	Main limitation
in situ TEM	Transmitted-electron image, optionally EDS/EELS	How does a visible structure evolve under bias?	Thin specimen, beam and vacuum may perturb it.
STM/STS	Tunneling current or differential conductance	How does a local surface electronic response change?	Surface, tip and feedback influence the signal.
Conductive AFM	Contact current and topography	Where does current enter an accessible surface?	Contact and scan can alter the sample.
Scalpel SPM / PMCM	Sequential contact maps / force-modulated electrical response	What depth structure or mechanically sensitive region is inferred?	Reconstruction or pressure introduces additional assumptions.

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Link the device and the instrument



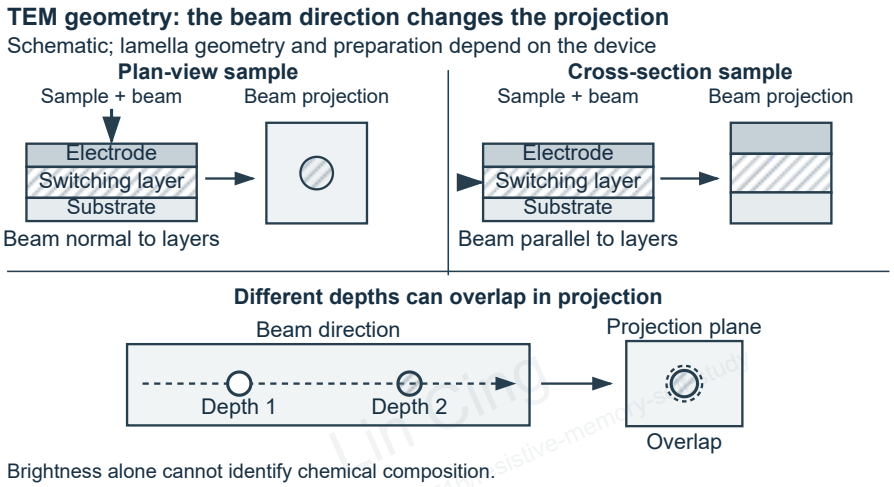
Record geometry, electrode assignments, voltage polarity, compliance, timing, environment and prehistory. The probe's electrical boundary is part of the experiment: a tip electrode is not automatically equivalent to an intact device's top electrode.

In situ TEM: what is actually observed?

An electron beam passes through a sufficiently thin specimen. The image is a **projection** of electron interactions through its thickness, not a direct three-dimensional concentration map. Biasing the specimen while acquiring images allows the electrical trace and structural evolution to be compared in time.

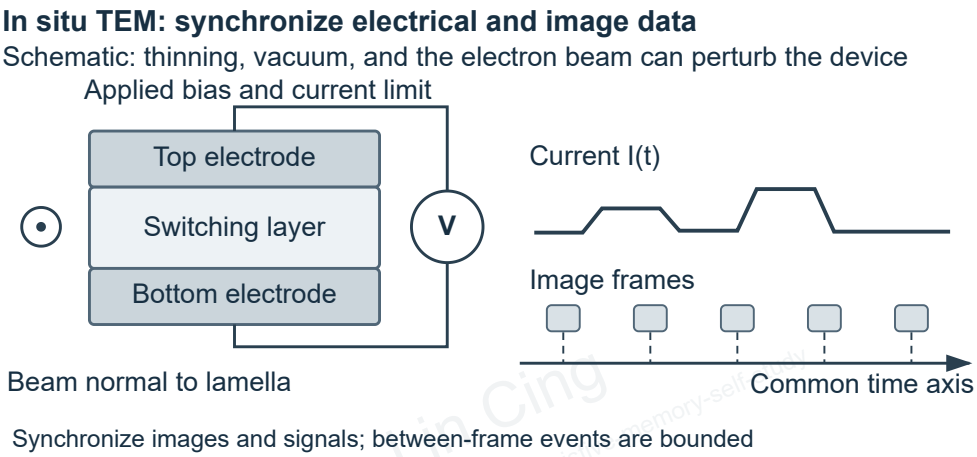
Different growth morphologies and nucleation positions are possible in different materials. A particular device's polarity or shape does not fix the direction for every ECM cell.

Cross-section and plan view: projections can overlap



In a cross-section view, objects separated along the beam can overlap as one feature in projection. For a deliberately ideal linear projection, two contributions of 2 and 3 arbitrary signal units sum to 5; that observed value alone cannot identify how many objects produced it. Real TEM contrast may be nonlinear and is not generally an additive concentration measurement. A plan-view projection looks from another orientation but still integrates along its own beam direction; tilting and matched electrical/chemical evidence help resolve the ambiguity.

Synchronize image and electrical trace



Synchronize images and signals; between-frame events are bounded

Before interpreting a feature, document which event is aligned with each frame, the exposure and clock uncertainty, and whether the device still switches after thinning. A frame is not necessarily an instantaneous sample: an exposure can average changes occurring during its interval.

A time-bound example

Suppose ideal instantaneous images are acquired every 30 ms. Frame 1 occurs at $t = 0$; frame 3 at 60 ms has no visible feature and frame 4 at 90 ms has one.

Under the ideal assumption that the feature becomes detectable once and remains detectable, its first detectable appearance is in $(60, 90]$ ms. The sequence alone does **not** give its exact onset, a speed or a chemical identity. Finite exposure, synchronization error and detection limits widen or complicate this inference.

Contrast, composition and oxidation state

Image contrast depends on specimen geometry, thickness, crystal orientation and imaging conditions as well as composition. A bright or dark region is evidence of contrast, not by itself a measurement of Ag, Cu or oxygen-vacancy concentration.

EDS provides elemental spectral information; EELS can provide elemental and chemical-state information when energy resolution, background and reference spectra support it. Both average over a finite interaction volume and can change the dose applied to the sample.

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TEM controls make a mechanism more credible

Compare an unbiased or lower-dose control, multiple switching cycles, and electrical behavior before and after imaging. Examine whether a morphology change coincides with a state change and whether composition-sensitive measurements support the proposed material.

FIB preparation, exposed surfaces, vacuum and the beam can affect oxygen content, interfaces, heating or damage. A non-switching thinned specimen cannot be silently treated as an undisturbed working device.

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Apparent motion has more than one explanation

Consider a cluster seen at position A in one frame and position B in a later frame. Translation is one hypothesis; dissolution near A and nucleation near B is another. Motion between frames or below resolution is not directly observed.

Vary time resolution and operating conditions, compare repeated trajectories and seek chemical or structural evidence. Report a mechanism as a supported model, not as the only possible explanation, when alternatives remain.

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STM: current across a tunneling gap

A conducting tip approaches a surface without the normal mechanical contact used in contact-mode AFM. At a fixed bias and small gap, the tunneling signal is strongly gap-dependent. Local surface states, tip states, the bias convention and the environment also influence current.

A pristine insulating film may not provide a stable measurable tunneling path; interpreting an image requires a specified sample stack and accessible surface.

In a simplified fixed-barrier model, $I \propto e^{-2\kappa z}$. If $\kappa = 10\,\mathrm{nm}^{-1}$ and the gap increases by 0.10 nm with all other parameters held fixed, $I_2/I_1 = e^{-2} \approx 0.135$. This is a model example, not a universal STM calibration.

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Feedback changes the meaning of an image

In **constant-current STM**, a feedback loop adjusts tip height to maintain a current setpoint while scanning. The recorded height reflects both geometry and local electronic response; a higher trace is not necessarily a physical protrusion.

In spectroscopy, an I–V trace and dI/dV depend on position, tip, bias, temperature and measurement protocol. Differential conductance may inform electronic states under a stated model; it is not an uncalibrated elemental map.

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A local write-and-read control

1. Image a region at a read condition chosen to minimize disturbance, and choose an unprogrammed control region.
2. Apply a specified local write pulse; record tip, pulse polarity, amplitude and duration.
3. Repeat low-disturb read at the same positions, then read again after a specified wait.
4. Compare current, topography and control drift. A persistent difference supports a retained local change under this protocol; it does not alone identify its chemical origin.

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Conductive AFM: contact current and topography

CAFM uses a conducting tip in mechanical contact with an accessible surface. A bias applied between the tip and another electrode yields a local current while the feedback system records topography. Tip force, coating, contact area, humidity, bias and current limit all affect the result.

The current map shows the response of **tip + contact + exposed sample + external circuit**, not an isolated, geometry-free property of the oxide.

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Two CAFM experiments with different boundaries

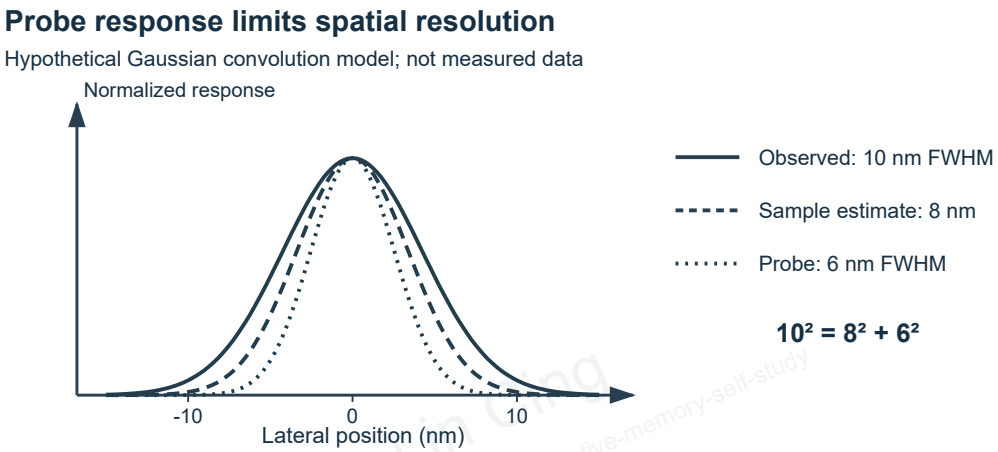
Tip-as-electrode: write and read directly on exposed film. The contact's field and force are part of the writing process.

Program first, expose later: switch a complete device electrically, remove its top electrode and map the exposed layer. Removal can damage or oxidize the feature, and the device cannot necessarily be cycled in the original configuration afterward.

Do not pool the two as if they had identical contacts, fields and sample histories.

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A current hotspot is not a direct diameter measurement



Requires stable linear Gaussian responses; hotspot width is not filament diameter

Measured width depends on the true spatial response, probe/contact geometry, sampling, feedback and noise. A visually wider hotspot can result from either a wider conducting region or a broader instrument response. Fit a stated forward model and show its limitations before inferring a size.

Gaussian resolution model: work through the numbers

Assume the true one-dimensional signal and a linear, shift-invariant point-spread function are Gaussian, with no additional broadening. Gaussian variances add; because $\mathrm{FWHM} = 2\sqrt{2\ln 2}\,\sigma$,

$$w_{\mathrm{obs}}^2 = w_{\mathrm{sample}}^2 + w_{\mathrm{tip}}^2.$$

For $w_{\mathrm{obs}} = 10$ nm and $w_{\mathrm{tip}} = 6$ nm, $w_{\mathrm{sample}} = \sqrt{10^2 - 6^2} = 8$ nm. The result is a model-dependent effective width, **not** a universal CAFM diameter correction.

If the two FWHMs have independent standard uncertainties of 0.5 nm, first-order uncertainty propagation gives $\sigma_w = \sqrt{(10 \cdot 0.5)^2 + (6 \cdot 0.5)^2}/8 \approx 0.73$ nm. This does not include uncertainty from model mismatch.

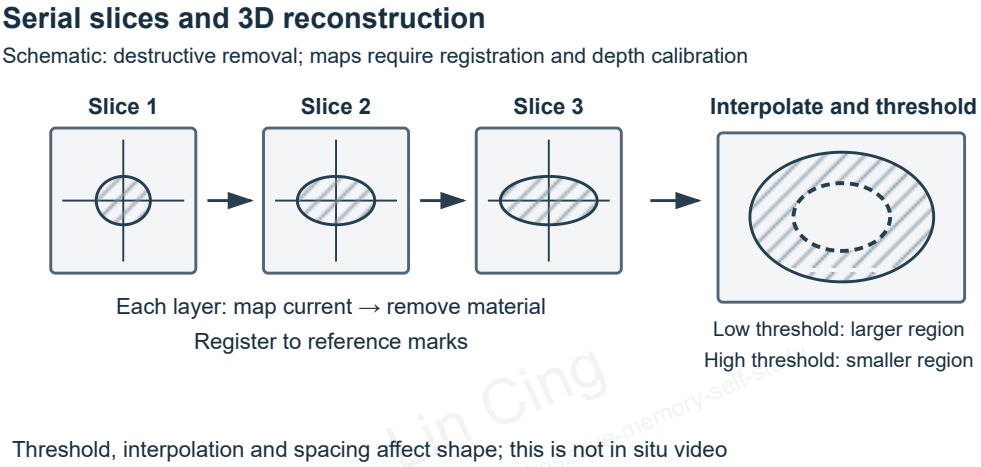
When deconvolution is not justified

If the measured width is no greater than the independently calibrated instrumental width, this Gaussian subtraction yields no positive real width. Within measurement uncertainty the feature may be unresolved; report an upper bound or a measurement limit rather than a fabricated diameter.

Even when subtraction is real, nonlinear contact, changing tip shape, non-Gaussian profiles and unknown uncertainty invalidate or weaken the estimate. Inspect raw profiles and calibration data.

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Scalpel SPM: a sequence, not a live 3D view



Program and locate a structure, remove material in controlled increments, and acquire a current map at each exposed surface. Register slices to a common reference and calibrate depth. The measurement is destructive and sequential; the sample may evolve between slices.

Reconstruction introduces choices

Spatial registration, depth estimates, interpolation, smoothing and the threshold used to draw an isosurface all affect the displayed shape. A continuous rendering is a visualization of measured slices **plus assumptions**, not independently measured data at every voxel.

Vary registration and threshold, use a stable reference electrode or separately characterized reference region at each slice to monitor tip changes, and report the slice spacing and removal uncertainty. A changing reference response flags possible tip wear or contamination; it cannot by itself rule out all local contact changes. Samples may oxidize or be mechanically damaged during exposure.

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Interpolate explicitly

At one registered lateral position, suppose measured currents are 2, 6 and 10 nA at depths $z = 0, 2, 4$ nm. With linear interpolation between the first two measurements,

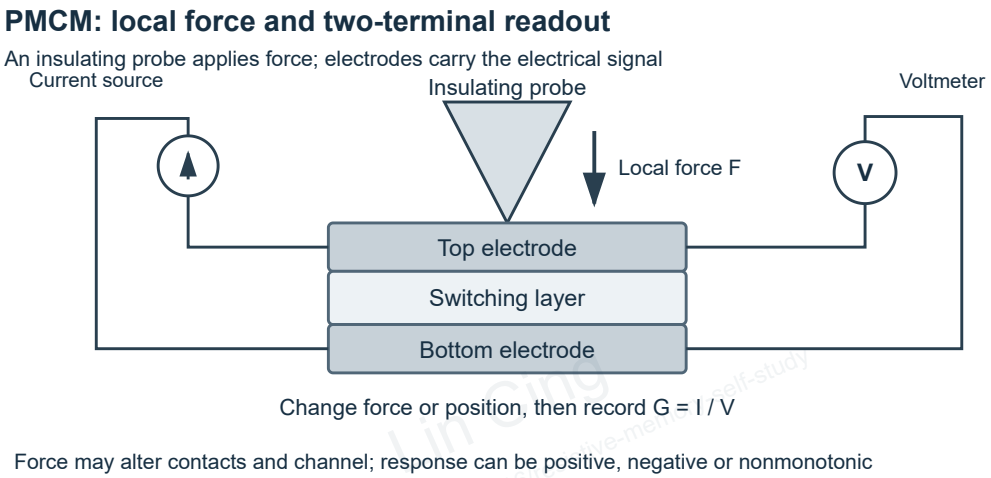
$$I(1\,\mathrm{nm}) = 2 + \frac{1-0}{2-0}(6-2) = 4\,\mathrm{nA}.$$

This 4 nA value is a model estimate, not a fourth measurement. An isosurface threshold of 3 nA includes different locations from one at 7 nA; a changing displayed volume is not by itself proof that the physical channel changed.

Similarly, interpolation between $z = 2$ and 4 nm gives $I(3) = 8$ nA and places the 7 nA crossing at $z = 2.5$ nm; both depend on the linear assumption.

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PMCM: a mechanical perturbation, an electrical readout



A non-conducting tip applies localized mechanical force to the top electrode of a complete device while the voltage across it is monitored under a low fixed current. Scanning the tip can reveal positions at which the whole-device resistance responds to that perturbation.

Force magnitude, load distribution, electrode mechanics, electrical bias, tip wear and spatial registration must be reported. The readout is not simply the current through the tip.

Calculate the conductance response

At fixed $I = 1\,\mu$A, suppose conductance rises from $G_0 = 10\,\mu$S to $G_1 = 12\,\mu$S when pressure is applied. Then

$$R_0 = 100\,\mathrm{k\Omega}, \quad R_1 = 83.33\,\mathrm{k\Omega}, \quad V_0 = 0.100\,\mathrm{V}, \quad V_1 \approx 0.0833\,\mathrm{V}.$$

The fractional conductance change is $(G_1 - G_0)/G_0 = 0.20$, or $+20\%$. A voltage dip at fixed current corresponds to lower resistance in this example; response sign and spatial shape depend on the device and mechanism.

What a PMCM contrast does and does not show

A repeatable force-correlated response identifies a mechanically sensitive region under specified conditions. It can guide cross-sectional observation, but does not by itself establish that every current path is there, reveal a unique filament geometry, or prove that conductance must rise under pressure.

Tunneling distances, electronic structure, mechanical deformation and quantum interference can contribute differently. Compare reverse scans, force-dependence and controls with other imaging and electrical measurements.

Lin Cing
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Coherent transport: why the response may change sign

For a phase-coherent conductor at zero temperature and infinitesimal bias, the Landauer expression is $G = G_Q \sum_n T_n(E_F)$, where $G_Q = 2e^2/h$ assumes spin degeneracy and T_n is the transmission probability of mode n. At finite temperature the energy-dependent transmission is weighted over the thermal window; finite bias requires an appropriate non-linear transport treatment.

For one illustrative lossless symmetric double barrier with single-barrier probabilities $T_b = 2/3$ and $R_b = 1/3$, summing the coherent multiple-reflection amplitudes gives $T(\phi) = T_b^2/|1 - R_b e^{i\phi}|^2 = 1/[1 + 3\sin^2(\phi/2)]$. The phase ϕ includes propagation and reflection contributions. This describes a model, not a measured device.

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Two phase points, opposite pressure responses

Assume, only for this example, that compression changes phase by $+\pi/6$ without changing the barriers. At $\phi = \pi/2$, T changes from $2/5$ to $4/13$; hence $\Delta G/G = -3/13 \approx -23.1\%$. At $\phi = 3\pi/2$, the same phase increment changes T from $2/5$ to $4/7$, giving $\Delta G/G = 3/7 \approx +42.9\%$.

The assumed phase shift is not a universal pressure law. Inelastic/dephasing processes, thermal averaging and several channels with different phases can weaken or alter the oscillation; other pressure-driven changes in barriers also violate the fixed-barrier model. An observed sign alone does not identify a unique mechanism.

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From observation to defensible claim

For every claim, attach (1) the measurement and sample preparation, (2) spatial and temporal resolution, (3) operating protocol and controls, and (4) competing interpretations.

Example: **Observed:** low-bias current map changes at a location after a pulse. **Supported:** a local electrical response changed under that protocol. **Not yet established:** that the spot is an isolated metal filament of the map's measured width. Composition and geometry require additional, matched evidence.

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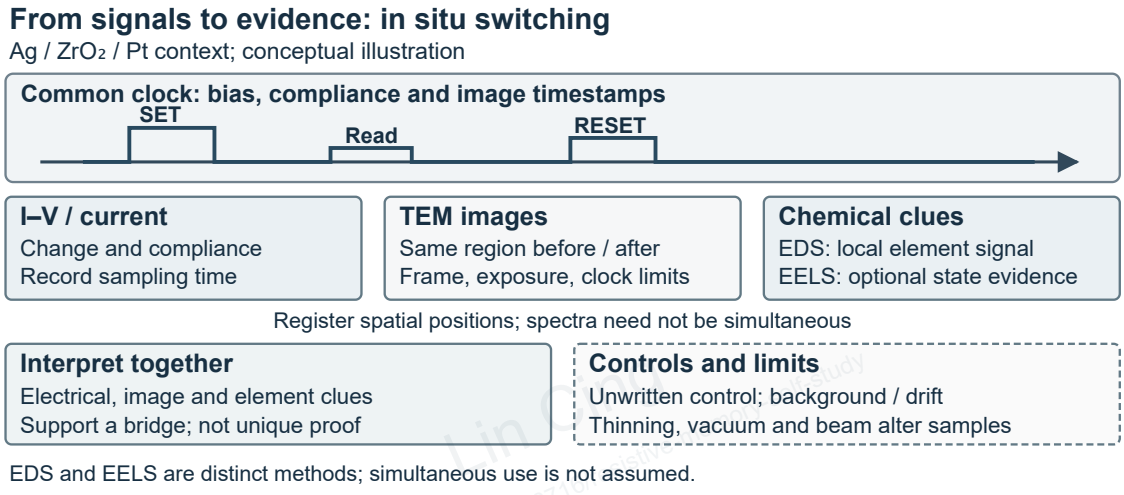
A practical experiment checklist

Stage	Questions to record
Before	Device stack and state; tip condition; alignment; environment; baseline trace.
During	Bias, compliance, current, force, acquisition time, clock synchronization and dose.
After	Repeatability, control regions, device survival, registration, uncertainty and alternative explanations.

Use the checklist to decide which follow-up experiment would distinguish two surviving mechanisms, rather than treating a visually persuasive image as a complete proof.

Lin Cing
https://github.com/xinhi6716/resistive-memory-self-study

A device-specific TEM evidence chain



In the reported FIB-prepared Ag/ZrO_2/Pt cell (approximately 100 nm scale), TEM images, SET/RESET I–V and EDS were compared. Ag-rich formation and dissolution were reported (source PDF p. 107). This figure is a general schematic, not the specimen or its data.

Align observations without inventing time resolution

For a proposed SET event, associate each image or chemical acquisition with its **actual exposure interval**, electrical trace and clock uncertainty. A post-SET EDS spectrum cannot be treated as a continuously recorded elemental movie. EELS is a complementary method in other experiments, not an assumed simultaneous measurement in this Ag example.

An EDS signal supports elemental attribution only with suitable local background and references; image contrast alone does not. Preparation, beam and vacuum may perturb the specimen: compare switching before and after imaging and use appropriate controls.

In a separate Pt/MoO_x/TiN video example, the source reports a 30 ms frame interval during RESET (PDF p. 109). It does not provide an exact transition time from that interval alone. Several visually different growth and dissolution modes are reported for different stacks; the material, electrode, environment and protocol matter.

Compare TEM cases rather than a universal filament

Reported case	Observation under its experiment	Interpretation and remaining limit
Ag cluster in evaporated SiO_2, electric field applied	A cluster disappears while another emerges nearby.	Dissolution and re-nucleation can mimic translation; frames do not show all intermediate processes.
Cu/WO_x/TiN, 10 cycles with SET compliance increased from 20 to 300 µA	HRS degraded as the protocol and contrast evolved.	The authors associate the changes with Cu deposition; compliance and cycle history change together, so this sequence alone is not a controlled single-variable proof.
Pt-Ir/NiO/Pt-Ir, forming with a 1 kΩ series resistor	An oxygen-poor region formed; subsequent RESET was not achieved.	The report attributes failure to oxygen lost to vacuum; thinning, current overshoot and environment restrict generalization.

These are distinct devices and measurement protocols (source PDF pp. 104, 110–113); do not transfer a frame interval or current limit between rows.

A counted EDS peak: a deliberately simple example

Suppose two equally exposed spectra use the same collection geometry and calibration. The measured Ag-peak window has 80 counts after SET and 50 after RESET; the estimated background is 20 counts in each window. Net counts are $80 - 20 = 60$ and $50 - 20 = 30$: the first is twice the second.

These are invented counts for arithmetic practice. A net-count ratio is **not** automatically a concentration ratio: thickness, detector efficiency, overlap, statistical uncertainty and local geometry matter. Compare reference regions and trace the image/spectrum acquisition to the same bias history before attributing a state change to Ag motion.

Lin Cing
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CAFM: one sample, multiple forming locations

In one reported HfO_2/Zr/TiN sample, a PtIr-coated tip was used as the local top electrode at 12 selected positions. Two locations had a breakdown/forming voltage around 5 V and ten had values above 10 V; the reported subsequent switchable response appeared only at the two lower-voltage locations (source PDF p. 132).

For this sample and selection, $2/12 \approx 16.7\%$. It is not a device-yield specification or an estimate for every film: only 12 locations were tested, and tip geometry, local force, sampling, history and compliance affect the observation. Follow each location with a defined read/write/read sequence and an appropriate control.

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Scalpel SPM: identify which area is being reported

Example	Electrical preparation and display	What the maps support	Limit
A reported CBRAM cell	After device programming, the reported CBRAM sequence used ±2 V and 50 µA compliance; a 5 nA isosurface was displayed.	The reported displayed area changed from about 493 to 200 nm^2 across the oxide.	The authors estimated about 10% area overestimation for this particular analysis; threshold and tip contact affect the result.
A reported oxide-vacancy switching cell	A separate example used ±1 V and a 50 pA display threshold.	Sequential conductive maps help infer a depth-dependent feature.	Exposed oxide may re-oxidize; a different display threshold is not evidence of the same physical area.

Both reconstructions use sequential removal rather than simultaneous 3D imaging (source PDF pp. 140–142). The thresholds above are **display choices**, not forming currents or universal geometric boundaries.

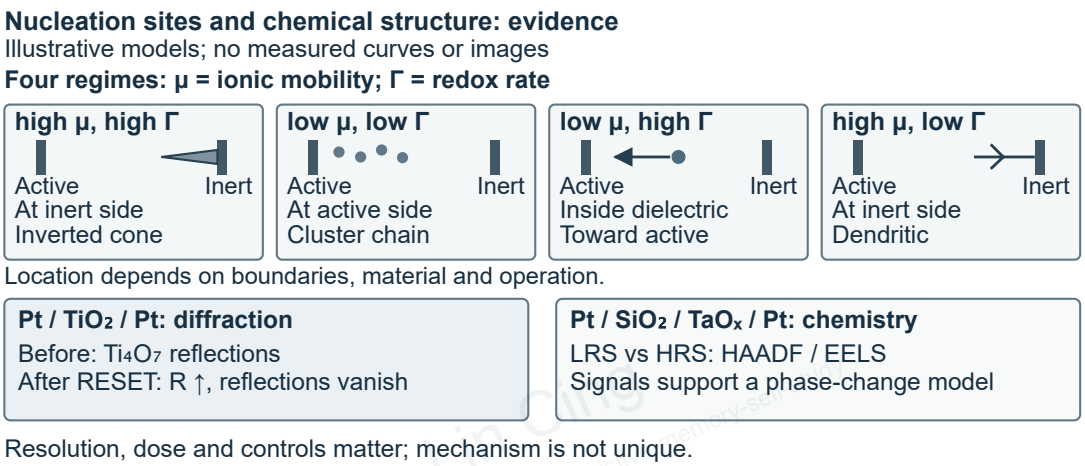
Four qualitative growth regimes

The source relates preferred nucleation and growth to ion mobility and redox rate (PDF pp. 98–102). “High” and “low” are qualitative within this model, not universal numerical thresholds.

Ion mobility	Redox rate	Model picture and reported illustration
High	High	Inert-electrode nucleation; tapering bridge. Example: $Cu/Al_2O_3/Pt$.
Low	Low	Near active source; separated clusters. Example: Ag/a-Si device.
Low	High	Interior nucleus growing back toward source. Example: Ni in evaporated SiO_2.
High	Low	Inert-electrode nucleation and branches. Example: Ag in sputtered SiO_2.

Geometry, gradients, history and detection also matter: polarity or shape alone does not determine the mechanism.

Diffraction and EELS: two complementary case studies



In a reported SET Pt/TiO_2/Pt specimen containing a conductive Ti_4O_7 Magnéli-phase feature, in situ RESET increased resistance while its superlattice diffraction reflections disappeared (source PDF p. 113). Correlated diffraction and electrical change support a structural contribution; they do not on their own determine the full three-dimensional path or exclude every accompanying chemical change.

Read chemical-state evidence with its controls

In a **separate** Pt/SiO_2/a-Ta_2O_{5-x}/a-TaO_{2-x}/Pt sample, the reported bipolar experiment achieved an on/off resistance ratio above 100. The authors compared HRS and LRS HAADF-STEM images and STEM-EELS spectra; for their specimen, they interpreted the filament's LRS as metallic TaO and the HRS as an insulating mixed-valence Ta_2O_5–TaO_2 region (source PDF pp. 114–115). These are local chemical-state interpretations under their analysis, not unique assignments established by a spectrum alone or universal claims for all TaO_x cells.

Compare the spectral energy calibration, reference spectra, background and spatial registration; assess whether thickness, beam dose or sample changes could affect the signal. A different diffraction pattern or EELS spectrum supports a state-dependent structural or chemical difference, but a contrast or spectrum alone need not uniquely establish the conducting path.

References and attribution

Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

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